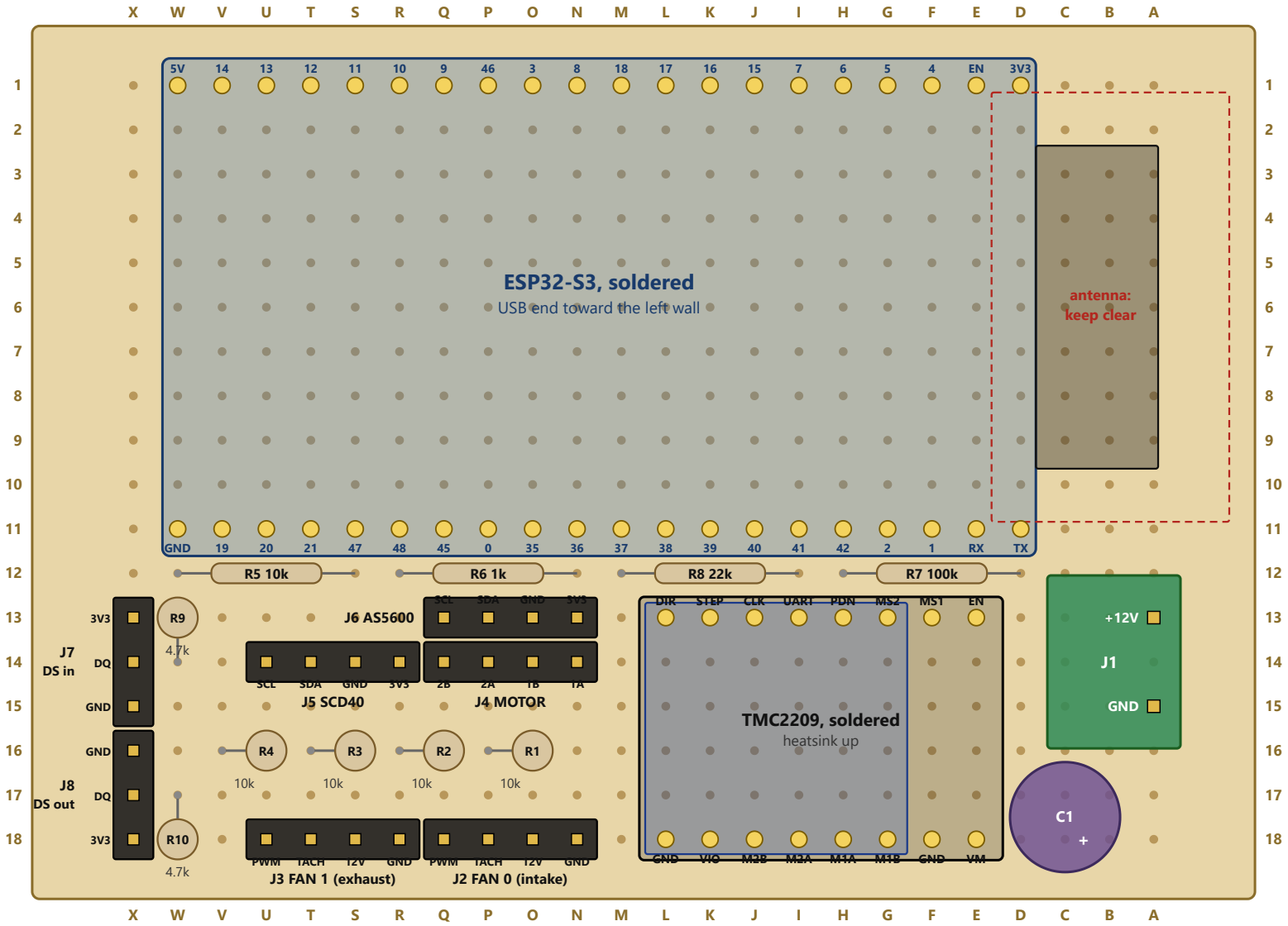
Perfboard build plan (5 x 7 cm protoboard, 24 x 18 holes)

Columns A to X from the right wall, rows 1 to 18 from the back (A1 is the board's back right corner, seen from above).

Page 1 of 3. Component side, seen from above (the room at the bottom)



Values: R1-R5 10k, R6 1k, R7 100k, R8 22k, R9 R10 4.7k (1/4 W); C1 100 uF 25 V. R1-R4, R9, R10 stand up; R5-R8 lie flat in row 12, so they pass under the edges of the ESP32 and the TMC2209 (2.6 mm up on their pin spacers).

Headers: 1 x 4 and 1 x 3 male, 2.54 mm. The fan plugs (J2, J3) go on with their latch toward row 17; keep row 17 clear there.

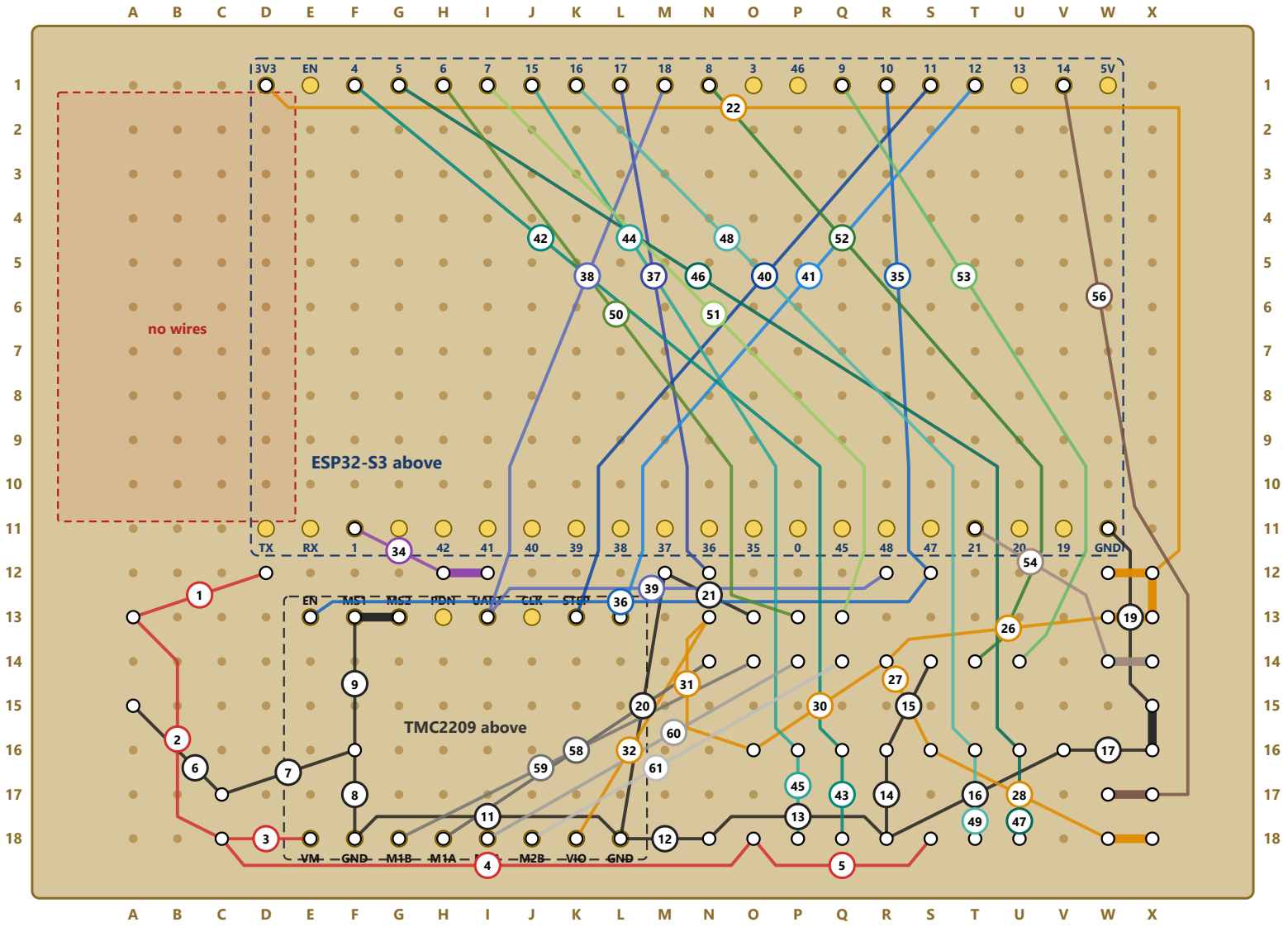
AS5600: tie its DIR pin to GND at the module. TMC2209: CLK and PDN stay free (wires 38-39 go to its UART pin; check with a meter that UART is PDN_UART on your module); DIAG (GPIO 13) is not on its pins, so it stays free too.

Solder the parts first, then the ESP32 and the TMC2209: the wires under them must be in place before.

Check each net with a meter before power; 12 V on J1 only with the motor plugged in.

Perfboard build plan (5 x 7 cm protoboard, 24 x 18 holes)

Page 2 of 3. Solder side: the board turned over left to right (A now on the left). Numbered wires as in the table.



Lines are the wires on the underside, coloured by net; the numbers match the list on page 3. Thick short bars are solder bridges between neighbouring holes. Dashed outlines: the ESP32 and the TMC2209, on the other side.

Perfboard build plan (5 x 7 cm protoboard, 24 x 18 holes)

Page 3 of 3. Wires (solder side), with the numbers of page 2

Thin insulated wire (wire-wrap or 26-30 AWG), except 22-24 AWG for the 12 V and GND around the driver (1-14) and the motor (58-61).

1	12V	A13 J1 +12V	to	D12 R7
2	12V	A13 J1 +12V	to	C18 C1 +
3	12V	C18 C1 +	to	E18 TMC VM
4	12V	C18 C1 +	to	O18 J2 12V
5	12V	O18 J2 12V	to	S18 J3 12V
6	GND	A15 J1 GND	to	C17 C1 -
7	GND	C17 C1 -	to	F16 tie point
8	GND	F18 TMC GND	to	F16 tie point
9	GND	F16 tie point	to	F13 TMC MS1
10	GND	F13 TMC MS1	to	G13 TMC MS2 (bridge)
11	GND	F18 TMC GND	to	L18 TMC GND
12	GND	L18 TMC GND	to	N18 J2 GND
13	GND	N18 J2 GND	to	R18 J3 GND
14	GND	R18 J3 GND	to	R16 R2
15	GND	R16 R2	to	S14 J5 GND
16	GND	R18 J3 GND	to	V16 R4
17	GND	V16 R4	to	X16 J8 GND
18	GND	X16 J8 GND	to	X15 J7 GND (bridge)
19	GND	W11 ESP GND	to	X15 J7 GND
20	GND	M12 R8	to	L18 TMC GND
21	GND	O13 J6 GND	to	M12 R8
22	3V3	D1 ESP 3V3	to	X12 tie point
23	3V3	X12 tie point	to	W12 R5 (bridge)
24	3V3	X12 tie point	to	X13 J7 3V3 (bridge)
25	3V3	X13 J7 3V3	to	W13 R9 (bridge)
26	3V3	W13 R9	to	R14 J5 3V3
27	3V3	R14 J5 3V3	to	S16 R3
28	3V3	S16 R3	to	W18 R10
29	3V3	W18 R10	to	X18 J8 3V3 (bridge)
30	3V3	R14 J5 3V3	to	O16 R1
31	3V3	O16 R1	to	N13 J6 3V3
32	3V3	N13 J6 3V3	to	K18 TMC VIO
33	SENSE	H12 R7	to	I12 R8 (bridge)
34	SENSE	H12 R7	to	F11 ESP GPIO 1
35	EN	R1 ESP GPIO 10	to	S12 R5
36	EN	S12 R5	to	E13 TMC EN
37	TX	L1 ESP GPIO 17	to	N12 R6
38	UART	M1 ESP GPIO 18	to	I13 TMC UART
39	UART	R12 R6	to	I13 TMC UART
40	STEP	S1 ESP GPIO 11	to	K13 TMC STEP
41	DIR	T1 ESP GPIO 12	to	L13 TMC DIR
42	PWM0	F1 ESP GPIO 4	to	Q16 R2
43	PWM0	Q16 R2	to	Q18 J2 PWM
44	TACH0	J1 ESP GPIO 15	to	P16 R1
45	TACH0	P16 R1	to	P18 J2 TACH
46	PWM1	G1 ESP GPIO 5	to	U16 R4
47	PWM1	U16 R4	to	U18 J3 PWM
48	TACH1	K1 ESP GPIO 16	to	T16 R3
49	TACH1	T16 R3	to	T18 J3 TACH
50	SDA1	H1 ESP GPIO 6	to	P13 J6 SDA
51	SCL1	I1 ESP GPIO 7	to	Q13 J6 SCL
52	SDA0	N1 ESP GPIO 8	to	T14 J5 SDA
53	SCL0	Q1 ESP GPIO 9	to	U14 J5 SCL
54	DQi	T11 ESP GPIO 21	to	W14 R9
55	DQi	W14 R9	to	X14 J7 DQ (bridge)
56	DQo	V1 ESP GPIO 14	to	X17 J8 DQ
57	DQo	X17 J8 DQ	to	W17 R10 (bridge)
58	M1A	H18 TMC M1A	to	N14 J4 1A
59	M1B	G18 TMC M1B	to	O14 J4 1B
60	M2A	I18 TMC M2A	to	P14 J4 2A
61	M2B	J18 TMC M2B	to	Q14 J4 2B